

PIT WALL — Prior Art & Reference Implementation Sweep

IBM SkillsBuild AI Builders Challenge | PhysicsTTM Architecture

1. Frozen Foundation Model + Physics-Projection Architecture

The direct question: Has anyone published an architecture that takes a FROZEN, pretrained time-series foundation model and wraps it with a physics-constraint or physics-projection layer *at inference time*, without retraining the model?

What Exists (Adjacent / Partially Relevant)

- [EXISTS — ADJACENT] "Learning Physical Constraints with Neural Projections" — Shuqi Yang, Xingzhe He, Bo Zhu. *NeurIPS 2020*. [Paper](#). This is the closest conceptual predecessor. A *neural projection operator* is used to enforce physical constraints on the output of a neural predictor of physical dynamics. The operator is motivated by position-based dynamics and learns to enforce constraints (length, angle, bending, collision, boundary) without explicit supervision. Crucial difference from PhysicsTTM: The predictor network is trained *together* with the projection operator end-to-end, not applied to a frozen, domain-agnostic foundation model. The constraint layer is lightweight and sits on the outside of the predictor, which is the same architectural slot as Layer B in PhysicsTTM.[¹][2]
- [EXISTS — ADJACENT] "PINT: Physics-Informed Neural Time Series Models" — ICLR 2026 (presented 2025). Introduces a framework for integrating physics constraints into neural time-series models (RNN, LSTM, GRU). Applied to ERA5 WeatherBench temperature forecasting using a Simple Harmonic Oscillator as a physics prior. Crucial difference: The physics is baked into the training objective; the model is not frozen, and the architecture is a custom-trained one.[³]
- [EXISTS — ADJACENT] "Zero-Shot Physics-Informed Fine-Tuning" — Survey/collection described at Emergent Mind (2026). Adapts *pretrained* models to new physical systems using only PDE residuals and boundary conditions as supervision, with LoRA for parameter-efficient adaptation. Crucial difference: The model weights *are updated* (even if via efficient PEFT); it is not a frozen inference-time wrapper. However, the paradigm of enforcing physics without labeled data on a pretrained backbone is closely related.[⁴]
- [EXISTS — ADJACENT] "Physics-Informed Diffusion Models for Vehicle Speed Trajectory Generation" — arXiv:2602.05028 (2026). Adds soft physics constraints to a diffusion model (1D U-Net and CSDI transformer) for driving profile generation. Tests on 6,367 GPS-derived micro-trips. Crucial difference: Model is trained with physics constraints in the loss; not a frozen foundation model.[⁵]

- [EXISTS — ADJACENT] "In-Context Fine-Tuning for Time-Series Foundation Models" — Matthew Faw, Abhimanyu Das, Ivan Kuznetsov, Rajat Sen, Yichen Zhou. *ICML 2025 / PMLR 2025*. Transforms TimesFM into a few-shot learner by prompting the frozen base model with in-context examples at inference time — no weight updates. This demonstrates that *inference-time wrapping* of a frozen TSFM is a live research area. Crucial difference: The adaptation mechanism is in-context examples (not physics projection), and there are no physical-validity guarantees.^{[6][7]}
- [EXISTS — ADJACENT] "Physics-Constrained Graph Neural Networks for Spatio-Temporal Prediction" — arXiv:2411.01848 (2024). Uses physics-constraint-based loss functions on a MeshGraphNet for simulation of physical systems. Not a frozen-model wrapper.^[^8]
- [EXISTS — ADJACENT] "Vehicle State Estimation combining PINN and UKF-M" — PMC 2023. Combines a PINN with an Unscented Kalman Filter on manifolds (UKF-M) to enforce kinematic constraints on vehicle state. The PINN and UKF-M are trained jointly, not wrapping a frozen model.^[^9]

The Confirmed Absence

- [CONFIRMED ABSENT] There is no published paper that takes a *frozen, pretrained, domain-agnostic time-series foundation model* (e.g., Granite TTM, Chronos, TimesFM, Moirai) and wraps it at inference time with a *hard differentiable physics-projection layer* that projects the output tensor onto a kinematically valid manifold — especially in a vehicle-dynamics context. Every prior-art paper either (a) trains the model together with the physics, (b) fine-tunes weights on physics-based losses, or (c) applies soft constraints only. The frozen-TSFM + hard-constraint QP projection pipeline (PhysicsTTM Layers $A \rightarrow B \rightarrow C$) with a Guardian audit layer is new architecture territory.

2. Differentiable Vehicle-Physics Code

Pacejka "Magic Formula" Tire Model

- [EXISTS] BrianN92/Pacejka-tire-model — GitHub. Python implementation of the full Pacejka MF tire model including lateral force (MagicFormula.py), self-aligning torque, normal force, and a lite lateral-only version. Reads .TIR files and YAML parameters. Pure NumPy; *not natively differentiable* but can be auto-differentiated via JAX or torch-wrapped. URL: <https://github.com/BrianN92/Pacejka-tire-model>^[^10]
- [EXISTS] mechmotum/Magic-Formula-89 — GitHub. MATLAB/Python implementation of Pacejka '89 (lateral force F_y and self-aligning torque M_z , 14 + 18 parameters). Pre-fitted optimal parameters available via Zenodo DOI. URL: <https://github.com/mechmotum/Magic-Formula-89>^[^11]
- [EXISTS] Data-Driven Differentiable Pacejka + Single-Track Model in PyTorch — OpenReview / PMLR (FITenth sim-to-real paper). A *fully differentiable* vehicle simulator coupling a single-track (bicycle) dynamics model with Pacejka MF6.1 tire forces and first-

order actuator dynamics is described in *"Data-Driven Simulation for Training High-Performance Racing Policies"* (F1TENTH). System identification is solved via backpropagation through time (BPTT) on 36 minutes of motion-capture data. This is the most direct proof-of-concept that differentiable Pacejka is both feasible and done in PyTorch.^[12]

- [EXISTS] `openVD (MATLAB/Octave)` — Documents full Pacejka Magic Formula implementation with equivalent slip angle formulation covering $\pm 180^\circ$ slip range. Python translation is available. URL: <https://andresmendes.github.io/openvd>^[13]

Single-Track and Double-Track Vehicle Models

- [EXISTS] `commonroad-vehicle-models (PyPI/pip)` — TU Munich Cyber-Physical Systems Group. Python implementations of: Kinematic Single-Track (KS), Linear Single-Track, Nonlinear Single-Track (ST), Double-Track, and Multi-Body models. Used as the backend for F1TENTH gym. `pip install commonroad-vehicle-models`. **Not natively differentiable** (uses SciPy ODE solvers), but models are pure Python/NumPy and can be re-expressed in PyTorch.^{[14][15]}
- [EXISTS] `TUMFTM/sim_vehicle_dynamics` — GitHub. Three variants of real-world racecar vehicle dynamics (non-linear single-track and dual-track models) in MATLAB/Simulink and Python. URL: https://github.com/TUMFTM/sim_vehicle_dynamics^[16]
- [EXISTS] `TUMFTM/YawMomentDiagrams` — GitHub. Python and MATLAB/Simulink implementations of non-linear single-track and dual-track models usable for GGV surface generation. URL: <https://github.com/TUMFTM/YawMomentDiagrams>^[17]
- [EXISTS] `TUMFTM/NeuralNetwork_for_VehicleDynamicsModeling` — GitHub. Neural network training algorithm to substitute a physics-based single-track bicycle model for vehicle dynamics — provides a PyTorch-differentiable *surrogate* model for the single-track equations.^[18]
- [EXISTS] `roadview-project/vehicle-dynamics` — GitHub. Python package for vehicle dynamics calculations (THI-CARISSMA and VTI). Includes optimization utilities over vehicle parameters.^[19]
- [EXISTS] `RudolfReiter/vehicle_gym` — GitHub. Python environment collecting functions for vehicle simulations: essential solvers, vehicle models, and Frenet transformation tools. URL: https://github.com/RudolfReiter/vehicle_gym^[20]
- [EXISTS] "Vehicle single track modeling using physics guided neural differential equations" — arXiv:2403.11648, 2024. Integrates a neural ODE network into the physical structure of a vehicle single-track model. Reduces sum-of-squared-error by 68% vs. pure physics model. **Not frozen-model inference**, but offers a differentiable hybrid single-track model.^[21]

Friction-Ellipse / Friction-Circle Constraints

- [EXISTS — Reference only, not differentiable] "The Tire-Force Ellipse (Friction Ellipse) and Tire Characteristics" — Brach & Brach, SAE 2011-01-0094 (2011). Canonical reference for the friction ellipse formulation $\left(\frac{F_x}{F_{x,max}}\right)^2 + \left(\frac{F_y}{F_{y,max}}\right)^2 \leq 1$.^[22]
- [EXISTS] commonroad-drivability-checker — TU Munich. Python source contains `FrictionCircleException` and a `VehicleDynamics` class with friction-circle feasibility checks built in. URL: <https://commonroad.in.tum.de>^[15]
- [EXISTS] TUMFTM/global_race_trajectory_optimization — GitHub. Uses friction maps and GGV surfaces (generalized friction ellipse) as constraints in minimum-time trajectory optimization. Differentiable-ish via CasADi (not PyTorch), but the constraint geometry is there.^[23]
- [CONFIRMED ABSENT — specifically differentiable PyTorch/JAX friction-ellipse QP layer] No pip-installable PyTorch/JAX module that expresses the friction ellipse as a differentiable hard constraint in a QP layer was found as a standalone library. The geometry is implementable using `CvxpyLayers` or `qpth` (see below), but no pre-packaged "FrictionEllipseLayer" exists.

Differentiable Optimization Layers

- [EXISTS] `cvxpy/cvxpylayers` — GitHub / PyPI. The canonical library for differentiable convex optimization layers in PyTorch, JAX, and TensorFlow. Wraps CVXPY problems as `CvxpyLayer` — solves in forward pass, differentiates via implicit differentiation in backward pass. Accompanies the NeurIPS 2019 paper "Differentiable Convex Optimization Layers" (Agrawal et al.). `pip install cvxpylayers[torch]`. URL: <https://github.com/cvxpy/cvxpylayers>^{[24][25][26]}
- [EXISTS] `locuslab/qpth` — GitHub / PyPI. Fast differentiable QP solver for PyTorch (primal-dual interior point). Accompanies "OptNet: Differentiable Optimization as a Layer in Neural Networks" (Amos & Kolter, ICML 2017). `pip install qpth`. URL: <https://github.com/locuslab/optnet>^{[27][28][29]}
- [EXISTS] `facebookresearch/theseus` — GitHub. Meta AI's application-agnostic library for differentiable nonlinear least squares (DNLS) optimization in PyTorch. Supports sparse solvers, automatic vectorization, batching, GPU acceleration, and implicit differentiation. For *nonlinear* constraints beyond QP. Presented at NeurIPS 2022 (Pineda et al.). URL: <https://github.com/facebookresearch/theseus>^{[30][31][32]}
- [EXISTS] `cwmagoon/dQP` — GitHub. Solver-agnostic, modular framework for differentiable QP ("dQP"), presented at NeurIPS 2025 (Magoon, Yang, Aigerman, Kovalsky). Decouples QP solution from differentiation; integrates with 15+ state-of-the-art solvers. Particularly strong on large-scale sparse problems. URL: <https://github.com/cwmagoon/dQP>^{[33][34][35][36]}
- [EXISTS] `kevin-tracy/qpax` — GitHub. Differentiable QP solver in JAX (primal-dual interior point). Works with `jit`, `vmap`, and `jax.grad`. `pip install qpax`. URL: <https://github.com/kevin-tracy/qpax>^[37]

- [EXISTS] "Differentiable Convex Optimization Layers in Neural Architectures: Foundations and Perspectives" — Katyal (Yale), arXiv:2412.20679 (2024). Comprehensive survey of the field from early QP implementations to general convex optimization layers. Essential reading for Layer B architecture choices.[⁴³]
- [EXISTS] Φ Flow (PhiFlow) — Holl & Thuerey, *ICML 2024*. Python toolkit integrating differentiable simulations (fluid PDEs, ODEs on grids) seamlessly with PyTorch, TensorFlow, JAX, and NumPy. Supports differential operators, boundary conditions, differentiable linear solves. Applicable to implementing differentiable Forward-Euler integration with hard constraints. URL: <https://github.com/tum-pbs/PhiFlow>[³⁹]

3. Race-Car Physics Simulators & Telemetry Tooling

Open-Source Lap-Time / Race-Car Simulators

- [EXISTS] mc12027/OpenLAP-Lap-Time-Simulator — GitHub / MATLAB Central. Open-source lap-time simulator in MATLAB. Includes OpenVEHICLE (inertia, engine torque curve, drivetrain, tyres, aerodynamics, steering), OpenTRACK (track geometry), and distance-based point-mass simulation. GPL-3.0 license. URL: <https://github.com/mc12027/OpenLAP-Lap-Time-Simulator>[⁴⁰][⁴¹]
- [EXISTS] TUMFTM/laptime-simulation — GitHub. Quasi-steady-state lap-time simulation in Python (TU Munich). Includes minimum-curvature raceline optimization, energy consumption modeling, vehicle parameter sensitivity. URL: <https://github.com/TUMFTM/laptime-simulation>[⁴²]
- [EXISTS] TUMFTM/global_racetrajectory_optimization — GitHub. Python algorithms for optimal racing line (shortest path, minimum curvature, minimum time). Includes friction maps, powertrain thermal model, GGV surface generation. URL: https://github.com/TUMFTM/global_racetrajectory_optimization[²³]
- [EXISTS] F1TENTH Gym — UPenn. Asynchronous Python 2D racing simulator running up to 30× realtime. Uses commonroad-vehicle-models single-track model with Pacejka-like tire forces. URL: https://github.com/f1tenth/f1tenth_gym[⁴³][¹²]
- [EXISTS] TUMFTM/Open-Car-Dynamics — GitHub. C++ implementation of vehicle dynamics (vehicle + drivetrain + steering actuator + aerodynamics + communication delays) for closed-loop simulation. URL: <https://github.com/TUMFTM/Open-Car-Dynamics>[⁴⁴]

FastF1 Python Library — Detailed Channel & Quality Audit

[EXISTS] the0ehrly/Fast-F1 — GitHub / PyPI. The principal Python library for accessing and analyzing Formula 1 results, schedules, timing data, and telemetry. URL: <https://github.com/the0ehrly/Fast-F1>[⁴⁵]

Car data channels (from F1 live timing API):[⁴⁶][⁴⁷][⁴⁸]

Channel	Type	Unit	Notes
Speed	float	km/h	Cars speed
RPM	float	rev/min	Engine RPM
nGear	int	—	Gear number 1–8
Throttle	float	0–100%	Pedal pressure; value 104 = error/unavailable
Brake	bool	—	Applied (True) or not (False); binary only, no pressure value
DRS	int	0/1/2/3	DRS availability + activation state

Position data channels:[⁴⁷][46]

Channel	Type	Unit
X	float	1/10 m
Y	float	1/10 m
Z	float	1/10 m
Status	str	OnTrack / OffTrack

Computed channels (added by FastF1 library):

- Distance, DistanceToDriverAhead, Time, SessionTime, Date, Source flag ('car' | 'pos' | 'interpolated')

Sample rate: The public F1 timing API delivers car data at approximately ~10 Hz per the community consensus on r/F1Technical; some sources cite sub-sample intervals. The OpenF1 API (separate project) explicitly states 3.7 Hz for its car telemetry endpoint. At 300 km/h, 10 Hz yields ~8 m between data points — insufficient for precise braking-point analysis.[⁴⁹][50][⁵¹]

Known data-quality issues:[⁵²][50][⁴⁶]

1. Brake channel is boolean — no analog brake pressure. You cannot derive brake force magnitude from FastF1 alone.
2. Throttle value 104 occasionally appears as an artifact when the car is stationary.
3. Irregular sampling — timestamps are non-uniform; the library adds an 'interpolated' source flag when resampling.
4. No lateral/longitudinal acceleration channels — G-forces must be computed from the derivative of speed/position, which is noisy at 10 Hz.
5. No steering angle channel — critically absent for vehicle dynamics work.
6. Pit lane geometry not encoded in the API; pit entry/exit must be manually set per circuit.
7. 2026 data loading issues — GitHub issue #865 reports telemetry not loading for the first two free practices of the 2026 season.[⁵³]

4. Public Motorsport Telemetry Datasets

- [EXISTS] FastF1 / F1 Live Timing Archive — Via the0ehrly/Fast-F1, accesses the official F1 timing API JSON stream (<http://livetiming.formula1.com>). Covers all sessions from 2018 onward. Channels: Speed, RPM, Gear, Throttle (0–100%), Brake (bool), DRS, X/Y/Z position. ~10 Hz car data, ~3.7–10 Hz position data.^{[54][46]}^[^47]
- [EXISTS] OpenF1 API — <https://openf1.org>. Free, open-source, no-auth REST API. 18 endpoints. Car telemetry at 3.7 Hz: speed, throttle, brake, RPM, gear, DRS. Position data (X/Y/Z). Historical data from 2023 season onward. Community/personal use free tier: up to 3 req/s. Live data available via paid sponsorship tier.^{[55][51]}^[^49]
- [EXISTS] TracingInsights F1 Data Archives — <https://tracinginsights.com/data>. Curated F1 data via FastF1 and Ergast API, organized by year (1950–2026). Raw telemetry and CSV format. Available on GitHub and Hugging Face. No API key required.^[^56]
- [EXISTS] RACECAR Dataset (Indy Autonomous Challenge) — Kulkarni, Chrosniak, Ducote et al. arXiv:2306.03252, IROS 2023. First open dataset for full-scale high-speed autonomous racing. Multi-modal sensor data from 6 autonomous Indy race cars at up to 170 mph (273 kph). 27 sessions, 11 racing scenarios, over 6.5 hours of sensor data. Channels: multiple LiDARs, cameras, radars, high-precision RTK-GPS + IMU. ROS2 and nuScenes formats. GitHub: https://github.com/linklab-uva/RACECAR_DATA^{[57][58]}^{[59][60]}
- [EXISTS] TUMFTM Racetrack Database — <https://github.com/TUMFTM/racetrack-database>. Center lines (X/Y GPS), track widths, and race lines for 20+ F1 and DTM tracks worldwide. Not telemetry, but critical for track-referenced analysis.^{[61][62]}
- [EXISTS — Sim-racing/open data] Ergast Developer API (via jolpi.ca mirror, 2025) — Historical F1 lap times, race results, standings from 1950–2024. No car telemetry. Timing data only.
- [CONFIRMED ABSENT] No publicly available real-world dataset for: (a) karting telemetry at any sample rate with documented channels, (b) GT3/GT4 club racing telemetry, (c) Formula Ford/junior single-seater telemetry, (d) adaptive/disabled-driver race telemetry. The exact category of driver that PIT WALL serves has zero public telemetry datasets.

5. Competitor Physics

Track Titan

- [CONFIRMED ABSENT — detailed physics documentation] Track Titan raised a \$5M seed round (December 2025) and positions itself as a "Strava for motorsport" / AI coaching platform. The product analyzes braking points, racing lines, and throttle traces from sim-racing (iRacing, ACC, F1 games) and real-world data. Their November 2025 update introduced "Coaching Flows," described as running *"thousands of different scenarios"* via a *"proprietary simulation engine"* to evaluate mistakes down to the millisecond. No technical whitepaper, academic publication, blog post, patent, or conference talk describing the

underlying physics or vehicle model has been found. The physics are proprietary. No evidence of a differentiable model or any formal vehicle dynamics layer.^{[63][64][65][66]}

Trophi.ai

- [CONFIRMED ABSENT — physics documentation] Trophi.ai is an AI coaching app for sim racing (iRacing, ACC) and other games (Rocket League). Development involved Scott Mansell (Driver61) and input from Renault F1 Team, Porsche, and Ferrari per user reviews. A December 2025 livestream by the trophi.ai data science team described the platform's evolution from telemetry analysis to real-time voice coaching — discussing how it turns data into "actionable coaching moments" — but contained **no mention of any physics model, vehicle dynamics model, or constraint layer**. The platform appears to be LLM-over-telemetry (compare your inputs vs. reference), not physics-simulation-based.^{[67][68]}

Formula E / Google Cloud "Driver Agent"

- [EXISTS — LLM architecture, no physics model] Announced March 2025. Powered by Google Cloud Vertex AI + Gemini. Processes multimodal race data (lap times, speed, braking, acceleration, G-force, downforce, latitude/longitude, steering) from a Formula E *simulator*. Architecture: Race log → GCS → Cloud Run preprocessing (fastest-lap ID, BigQuery tables, heatmaps, telemetry graphs as static images) → Gemini with multimodal inputs (tables + images) → text/audio coaching. The system **compares driver telemetry against a professional Formula E reference lap. No physics model, vehicle dynamics engine, or constraint layer is used**. It is a multimodal LLM reasoning over telemetry data. No academic paper or technical whitepaper has been published beyond the Google Cloud marketing blog post.^{[69][70][71][72][73]}

AiDE (Sim Racing)

- [CONFIRMED ABSENT — physics documentation] AiDE (AI Driver Engineer) appeared on r/simracing (May 2025) as a setup suggestion tool that takes driver qualitative feel and suggests setup changes. No physics model documentation found.^[74]

General Observation

No AI driving-coaching competitor has published: (a) a formal vehicle dynamics model, (b) a physics-constraint layer, (c) a differentiable physics manifold projection, or (d) any treatment of adaptive/disabled-driver physics. The field is uniformly LLM-over-telemetry or telemetry-comparison systems without validated physical models.

6. IBM Granite TimeSeries TTM In the Wild

Model overview: TinyTimeMixer (TTM) / Granite TimeSeries is a compact (<1M parameter) pretrained time-series foundation model based on TSMixer architecture. Introduced at NeurIPS 2024. Pre-trained on ~250M (r1) to ~1 billion (r2.1) time-series samples from **publicly available datasets: energy, weather, retail, and financial data**. Supports zero-shot and few-shot multivariate forecasting. Available under Apache 2.0 license on Hugging Face ([ibm -](#)

granite/granite-timeseries-ttm-r1 and r2). Now importable into IBM Watsonx.ai.^{[75][76][77][78][79][80]}

Documented real-world applications found:

Application	Source	Physics / Sensor Type
ERCOT energy demand forecasting (Texas power grid)	Medium / Kyle Jones, 2025	Power grid time series
HVAC system simulation and energy optimization	GFT Engineering, Medium, 2025	Building IoT sensors
Retail sales forecasting (M5 competition dataset)	IBM Tutorial, 2025	Retail tabular
Zero-shot forecasting on Seoul Bike Sharing Demand (UCI)	Oracle AI + OCI blog, 2025	Urban mobility
Stock prices / trading volumes	IBM Watsonx docs	Financial
EKG / PSG medical time series	IBM Watsonx docs (mentioned as example)	Biomedical sensor
Temperature / seismographic data	IBM Watsonx docs (mentioned)	Environmental
Network performance measurements	IBM Watsonx docs (mentioned)	IT infrastructure

^{[81][82][83][84][85]}

[CONFIRMED ABSENT — vehicle / motorsport / high-frequency physical telemetry:

No documented real-world application of Granite TTM (r1 or r2) applied to: vehicle dynamics, race-car telemetry, IMU/accelerometer data, steering/brake/throttle time series, or any high-frequency (>1 Hz physical actuation) sensor data from a moving vehicle has been found in any blog post, paper, GitHub repository, or IBM documentation as of May 2026. The model's design explicitly targets "minutely to hourly resolutions" (the r2 README explicitly states this), which is why PIT WALL downsamples from 50 Hz to 1 Hz — a necessary pre-processing step with no published precedent in a motorsport context.^[75]

7. Adaptive / Hand-Control Motorsport Engineering

The MME / Team BRIT System

- [EXISTS] Team BRIT / MME Motorsport hand control system — World's most advanced disability racing hand controls, developed jointly by Team BRIT (UK) and MME Motorsport (Slovenia, Marko Mlakar). Uses **pneumatic, electronic, and hydraulic** technology. All functions (steering, brakes, clutch, throttle, gears) are controlled via hands-only using an adapted steering wheel. Presented to the FIA; Team BRIT's Al Locke became an expert on FIA's international disability working group. Available exclusively through Team BRIT from January 2023. URL: <https://teambrit.co.uk/hand-control-technology/>^{[87][88][89][90][91]}
- [EXISTS — engineering description with simultaneous control detail] "Behind MME & Team BRIT's hand control technology" — PMW Magazine, October 2017. Detailed write-up

of the system. Key statement on simultaneous inputs: *"The module also allows for much more complex inputs, such as simultaneous control. The system is able to take inputs such as holding full throttle while shifting, or holding full throttle and applying a little braking mid-corner."* The system priority logic: *"You can still brake and press throttle with foot. Whichever component is pressed more, takes a priority."* This confirms that the COA flag in PhysicsTTM (toggling the throttle $\times$ brake = 0 constraint) corresponds to a real, documented, technically verified behavior of the MME system.^{[89][90]}

- [EXISTS — commercial description] MME Motorsport product page — <https://www.mme-motorsport.com/en/products/hand-controls>. Lists "Simultaneous control — Holding full throttle while shifting or holding full throttle and applying a little braking in the corner" as a documented feature advantage.^[^90]
- [EXISTS] Resilience Racing Foundation / Texas A&M Engineering Partnership — Documents a US-based effort to develop adaptive hand controls for club-level racing, including a "double ring steering wheel" design integrating 360° throttle and brake. Uses an electronic throttle body and pneumatic/hydraulic actuator. References MME Motorsport's clutch actuator. URL: <https://www.resilienceracingfoundation.org/hand-control-innovation>^[^92]
- [EXISTS] Robert Wickens Hand Control System — Developed by Bosch, Pratt Miller Motorsports, GM, and DXDT for IMSA competition. Custom by-wire system allowing Wickens to compete in TCR-class racing after his 2018 IndyCar crash. Covered in Grassroots Motorsports (2023) and YouTube (2025).^{[93][94][^95]}
- [EXISTS] FIA Vehicle Adaptation Guidelines (DAA) — api.fia.com/sites/default/files/maj2023_daa_guidelines_a4_v4_web.pdf. FIA's official document on Driving Adaptations and Assistance (DAA), covering throttle management by hand lever — mechanically or electronically linked — and brake adaptation. Provides the regulatory framework from which COA flags derive.^[^96]

Vehicle-Dynamics Consequences of Simultaneous Brake + Throttle

- [CONFIRMED ABSENT — peer-reviewed paper] No published academic paper specifically analyzing the vehicle-dynamics consequences of simultaneous brake-and-throttle application as used by adaptive hand-control racing drivers was found. The engineering of the control *mechanism* is documented (Team BRIT/MME), but the downstream effects on:
 - Longitudinal load transfer during combined throttle + brake
 - Friction ellipse utilization with non-zero longitudinal force in both directions
 - Tire thermal loading under simultaneous actuation
 - Understeer/oversteer balance changes from combined inputs

...have not been characterized in any published motorsport engineering or vehicle dynamics paper. This is a genuine research gap.
- [EXISTS — adjacent, road vehicle context] "Vehicle Speed and Spacing Control via Coordinated Throttle and Brake Actuation" — Sciencedirect, Daimler-Benz Research, 1997. Describes coordinated engine + brake controllers for automated highway driving (non-

racing). Not directly applicable to the adaptive driver scenario but establishes the longitudinal dynamics of simultaneous actuation.[⁹⁷]

- [EXISTS — general vehicle dynamics reference] Weight transfer physics under braking (longitudinal: $\Delta W = (m \cdot a \cdot h) / L$) and combined longitudinal + lateral loading are covered in standard texts (e.g., *Vehicle Dynamics: Theory and Application*, Jazar). No text addresses the simultaneous throttle + brake case from a racing driver adaptation standpoint.[⁹⁸][⁹⁹]

WHITE SPACE

Where PIT WALL Is First

1. **Frozen TSFM + Hard Differentiable Physics Projection.** No paper has ever taken a *frozen, domain-agnostic* time-series foundation model (pre-trained on energy/weather/retail, knowing nothing about Newtonian mechanics) and constrained its output at inference time with a hard QP-solved differentiable physics-projection layer. Every prior work either trains the model with the physics, fine-tunes weights, or uses only soft constraints. PhysicsTTM Layers A→B→C is architecturally unreported.
2. **Kinetic Hallucination as a Named Failure Mode.** The specific failure of a multivariate foundation model producing kinematically impossible cross-channel combinations (e.g., 4G lateral acceleration with zero steering input) — what PIT WALL calls "kinetic hallucination" — has not been named, defined, characterized, or mitigated in any published literature. All prior physics-informed neural network work addresses a different failure mode (PDE residual violation during training).
3. **Granite TTM Applied to Vehicle / Racing Telemetry.** Not one documented application of IBM Granite TTM exists in vehicle dynamics, motorsport, or any high-frequency physical-sensor context. PIT WALL will be the first published use case.
4. **AI Race Engineering System for Adaptive Drivers with COA-Aware Physics.** No AI coaching product (Track Titan, trophi.ai, Formula E Driver Agent) has a physics model at all — let alone one that reads an FIA Certificate of Adaptations and switches constraint logic based on documented disability-specific driving behaviors (simultaneous brake + throttle). The intersection of LLM-based coaching, differentiable physics, and adaptive motorsport engineering is entirely unoccupied.
5. **Published Characterization of Simultaneous Brake+Throttle Dynamics in Adaptive Racing.** The vehicle-dynamics consequences (load transfer, friction-ellipse utilization, tire thermal effects) of the simultaneous brake-and-throttle inputs permitted by MME-type hand controls remain uncharacterized in any published paper. PIT WALL's COA flag mechanism, combined with future empirical telemetry analysis, positions it to generate the first published data on this phenomenon.

References

1. [Learning Physical Constraints with Neural Projections - NIPS papers](#) - Abstract. We propose a new family of neural networks to predict the behaviors of physical systems by...
2. [Learning Physical Constraints with Neural Projections](#) - NeurIPS 2020. Learning Physical Constraints with Neural Projections. Review 1. Summary and Contribut...
3. [PINT: Physics-Informed Neural Time Series Models ... - ICLR 2026](#) - This paper introduces PINT (Physics-Informed Neural Time Series Models), a novel framework designed ...
4. [Zero-Shot Physics-Informed Fine-Tuning - Emergent Mind](#) - Adapt neural models to new physics tasks without labeled data by enforcing governing equations and l...
5. [Physics-Informed Diffusion Models for Vehicle Speed Trajectory ...](#) - Their key advantages include stable training dynamics, the ability to naturally model complex multi...
6. [In-Context Fine-Tuning for Time-Series Foundation Models](#) - We design a pretrained foundation model that can be prompted (at inference time) with multiple time...
7. [In-Context Fine-Tuning for Time-Series Foundation Models](#) - We show that such a foundation model that uses in-context examples at inference time can obtain much...
8. [Physics-Constrained Graph Neural Networks for Spatio-Temporal ...](#) - (2023) developed two approaches to accelerate structural analysis from a spatio-temporal perspective...
9. [Vehicle State Estimation Combining Physics-Informed Neural ... - PMC](#) - This paper proposes a novel vehicle state estimation (VSE) method that combines a physics-informed n...
10. [GitHub - JyNing04/Pacejka-tire-model](#) - This repository contains scripts and data related to tire dynamics and vehicle configuration, utiliz...
11. [GitHub - mechmotum/Magic-Formula-89](#) - This script generates the parameters for Pacejka 89 model, both for lateral force F_y and self-aligni...
12. [Data-Driven Simulation for Training High-Performance ...](#)
13. [Tire models — OpenVD documentation](#) - Pacejka (Magic Formula)¶. Nonlinear relationship between tire lateral force ... The model implemente...
14. [commonroad-vehicle-models](#) - Implementation of vehicle models with varying abstraction levels ranging from kinematic single track...
15. [Source code for commonroad_dc.feasibility.vehicle_dynamics](#)
16. [TUM Roborace Team Software Stack - Vehicle Simulation · GitHub](#) - The following repository covers the vehicle dynamics simulation of a real-world racecar and can be u...
17. [Tool to generate yaw moment diagrams for vehicle ... - GitHub](#) - This repository contains algorithms to generate Yaw Moment Diagrams (YMDs) on basis of existing vehi...
18. [Neural Network for Vehicle Dynamics Modeling - GitHub](#) - This repository provides a neural network training algorithm which is able to substitute a physics-b...
19. [GitHub - roadview-project/vehicle-dynamics: Python Package that makes Vehicle Dynamics Calculation for cars.](#) - Python Package that makes Vehicle Dynamics Calculation for cars. - GitHub - roadview-project/vehicle...

20. [GitHub - RudolfReiter/vehicle_gym](#) - Contribute to RudolfReiter/vehicle_gym development by creating an account on GitHub.
21. [Vehicle single track modeling using physics guided neural ... - arXiv](#) - In this paper, we follow the physics guided modeling approach and integrate a neural differential eq...
22. [\[PDF\] The Tire-Force Ellipse \(Friction Ellipse\) and Tire Characteristics](#) - The g-g diagram takes the form of the lateral acceleration of a vehicle. (usually a racing vehicle) ...
23. [GitHub - TUMFTM/global_racetrajectory_optimization](#) - This repository contains algorithms that allow us to determine an optimal racing line on a race trac...
24. [cvxpy/cvxpylayers: Differentiable convex optimization layers - GitHub](#) - CVXPYlayers is a Python library for constructing differentiable convex optimization layers in PyTorch...
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26. [\[PDF\] Differentiable Convex Optimization Layers - Stanford University](#) - We implement our methodology in version 1.1 of CVXPY, a popular Python-embedded DSL for convex op...
27. [OptNet/qpth Example Notebook](#) - A gallery of the most interesting jupyter notebooks online.
28. [OptNet: - Brandon Amos](#)
29. [qpth](#) - A fast and differentiable QP solver for PyTorch.
30. [Theseus: A Library for Differentiable Nonlinear Optimization - Meta AI](#) - We present Theseus, an efficient application-agnostic open source library for differentiable nonlinear...
31. [A Library for Differentiable Nonlinear Optimization - Theseus - arXiv](#) - We present Theseus, an efficient application-agnostic open source library for differentiable nonlinear...
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36. [Differentiation Through Black-Box Quadratic Programming Solvers](#) - We introduce dQP, a modular, solver-agnostic framework that transforms any QP solver into a differentiable...
37. [gpax-solver/gpax: Differentiable QP solver in JAX - GitHub](#) - This solver can be combined with JAX's jit and vmap functionality, as well as differentiated with re...
38. [Differentiable Convex Optimization Layers in Neural Architectures](#) - In this survey, we focus on the development of convex optimization layers as a means to enforce hard...
39. [Differentiable Simulations for PyTorch, TensorFlow and Jax](#) - We present Φ Flow, a Python toolkit that seamlessly integrates with PyTorch, TensorFlow, Jax and NumPy...

40. [OpenLAP-Lap-Time-Simulator - File Exchange - MATLAB Central](#) - OpenLAP Lap Time Simulator. OpenVEHICLE Vehicle modeling: Includes inertia, engine torque curve, dri..
41. [GitHub - mc12027/OpenLAP-Lap-Time-Simulator](#) - Open source lap time simulator coded in MATLAB. Contribute to ... Lap time simulation using a simple...
42. [TUMFTM/laptime-simulation: This repository contains a ... - GitHub](#) - This repository contains a quasi-steady-state lap time simulation implemented in Python. It can be u...
43. [F1TENTH Gym Documentation](#)
44. [TUMFTM/Open-Car-Dynamics: C++ Implementation of a ... - GitHub](#) - Open Car Dynamics provides a comprehensive, modular, and highly efficient implementation of a vehicl...
45. [theOehrly/Fast-F1: FastF1 is a python package for ... - GitHub](#) - FastF1 is a python package for accessing and analyzing Formula 1 results, schedules, timing data and...
46. [Car Telemetry — FastF1 \(3.8.1\)](#)
47. [Telemetry - FastF1 - Mintlify](#)
48. [Source code for fastf1.core](#)
49. [OpenF1 API | The open source API for Formula 1 data](#) - The open source Formula 1 API. Get real-time access to live telemetry, lap times, weather, and race ...
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51. [Free API Directory - TheFreeAPI.com](#) - Get F1 car details, team info, and season stats with the OpenF1 Car Data API for apps, websites, and...
52. [Working with FastF1 + real-time telemetry — struggling with track mapping & pit zones - Working with FastF1 + real-time telemetry — struggling with track mapping & pit zones](#)
53. [Telemetry not loading of the first two free practices of 2026 #865](#) - The following code has been cobbled together over months, with much try-and-error along the way. Muc...
54. [Telemetry data source, where to get it? : r/F1Technical - Reddit](#) - There is only one public telemetry source wherein telemetry is put up right after a session on the F...
55. [OpenF1 API - Real-time and historical Formula 1 data - GitHub](#) - OpenF1 is a free and open-source API that offers real-time and historical Formula 1 data. ... car te...
56. [Formula 1 Data Archives & Telemetry - TracingInsights](#) - Access our comprehensive collection of Formula 1 data from 1950 to latest race. Download raw telemet...
57. [GitHub - OSSDC/RACECAR_DATA](#) - Contribute to OSSDC/RACECAR_DATA development by creating an account on GitHub.
58. [linklab-uva/RACECAR_DATA - GitHub](#) - The RACECAR dataset is compiled by contributions from several teams, all of whom competed in the ina...
59. [AWS Marketplace: RACECAR Dataset](#) - The RACECAR dataset is the first open dataset for full-scale and high-speed autonomous racing. Multi...
60. [RACECAR -- The Dataset for High-Speed Autonomous Racing - arXiv](#) - This paper describes the first open dataset for full-scale and high-speed autonomous racing. Multi-m...

61. TUMFTM/racetrack-database: This repository contains center lines (x - This repository contains center lines (x- and y-coordinates), track widths and race lines for over 2...
62. Open Source Software - Chair of Automotive Technology - Open Source Software for Autonomous Racing ; Datasets, <https://github.com/TUMFTM/racetrack-database> ...
63. AI sim-racing coach Track Titan raises \$5m to train the next ... - AI sim-racing coach Track Titan raises \$5m to train the next generation of drivers. London-based Tra...
64. Track Titan raises USD \$5m to power AI racing coaching - IT Brief UK - London start-up Track Titan raises USD \$5m seed round to scale its AI-powered racing coach for gamer...
65. London's Track Titan raises \$5M for 'Strava of motorsport' AI coaching - Track Titan aims to expand globally, embedding AI into steering wheels, pedals, and car dashboards a...
66. November 2025 Track Titan Update | Coaching Flows to Actually ... - Forget telemetry tools and other so-called AI coaching platforms for simracers. There is finally a w...
67. AI Simracing Coach | Trophi.ai im Test | Künstliche Intelligenz als Coach? | - AI Simracing Coach | Trophi AI im Test |

Trophi AI ist ein Simracing Coach für iRacing und ACC. In...

68. We Built AI Coaching for Games | trophi.ai - In this live stream, Scott and Rob from the trophi.ai data science team break down how our AI coachi...
69. 'Driver Agent' sees Formula E and Google Cloud work together to ... - The Driver Agent is aimed at levelling the playing field to grant access to advanced AI data – inclu...
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72. Formula E's AI equation: A new Driver Agent for the next generation ... - The Driver Agent is designed to analyze extensive multimodal data generated during racing and provid...
73. How Formula E uses Google Cloud generative AI - Formula E is able to provide a conversational interface for their drivers to answer specific questio...
74. So, we're making an AI race engineer : r/simracing - Reddit - It's called AiDE – short for Ai Driver Engineer – and the idea is pretty simple: You tell it how you...
75. README.md · ibm-granite/granite-timeseries-ttm-r2 at ... - We're on a journey to advance and democratize artificial intelligence through open source and open s...
76. IBM Improves Generative AI Forecasting Using Time, Not Just ... - TTM is a lightweight pre-trained time series foundation model (TSFM) for time-series forecasting bas...
77. Tiny Time Mixers (TTMs): Fast Pre-trained Models for Enhanced ... - To address this, we introduce Tiny Time Mixers (TTM), a compact model (starting from 1M parameters) ...
78. Custom foundation models import support for Granite Time ... - IBM has expanded its Custom Foundation Models feature to support Granite Time Series models.

79. [GitHub - ibm-granite/granite-tsfm: Foundation Models for Time Series](#) - Run the TTM tutorial in Google Colab, and quickly build a forecasting application with the pre-train...
80. [Tiny Time Mixers \(TTMs\): Fast Pre-trained Models for Enhanced...](#) - We introduce Tiny Time Mixers (TTM), a compact model (starting from 1M parameters) with effective tr...
81. [Forecasting Energy Demand with IBM Granite TinyTimeMixer](#) - Energy demand forecasting has always been a challenge, especially for large-scale grids like ERCOT. ...
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83. [Evaluating IBM's Granite TTMs and the Impact of Time-Series Foundational Models on Industry: - A New Approach to Time-series Data Prediction](#)
84. [Use the time series forecast API and IBM Granite ...](#) - Use the time series forecast API and IBM Granite time series models that are available from watsonx....
85. [Time Series Sales Forecasting with Granite](#) - This tutorial demonstrates how to use Granite Time Series Foundation Models for sales forecasting, i...
86. [Zero-Shot Time Series Forecasting Using Granite TTM on OCI](#) - Built for zero-shot forecasting, it's designed to deliver accurate predictions across a wide range o...
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94. [This is the unique hand-control setup Robert Wickens uses to win](#) - While most racers use their feet to work the clutch, brake and gas, Robert uses only his hands in hi...
95. [How Robert Wickens' Hand Control System Works - - YouTube](#) - He uses a custom hand control system that Bosch, Pratt Miller Motorsports, GM, and DXDT developed. #...
96. [\[PDF\] VEHICLE ADAPTATION GUIDELINES - FIA](#) - Throttle management by hand lever may be mechanically linked to the pedals or electronically linked ...
97. [Vehicle speed and spacing control via coordinated throttle and brake actuation](#)
98. [Weight Transfer in Vehicle Dynamics – What Really Happens ...](#) - When braking shifts weight forward, the front tires gain traction while the rear tires lose traction...

99. [\[PDF\] Vehicle Dynamics: Theory and Application - Part I](#) “One Dimensional Vehicle Dynamics,” presents forward vehicle dynamics, tire dynamics, and dri...